Practical 4b: E-Commerce PWA with Product Listing and Offline Cart

AIM: To develop an E-Commerce Progressive Web Application with product listing and offline cart functionality using Service Worker and local Storage.

PROGRAM CODE

```
<!DOCTYPE html>
<html lang="en">

<head>
   <meta charset="UTF-8">

   <meta name="viewport"
       content="width=device-width, initial-scale=1.0">

   <title>ShopPWA - E-Commerce</title>

   <link rel="manifest" href="manifest.json">

   <meta name="theme-color" content="#E74C3C">

   <style>
      * {
         box-sizing: border-box;
      }

      body {
         font-family: Arial, sans-serif;
         margin: 0;
         background: #f5f5f5;
      }

      header {
         background: #E74C3C;
         color: white;
         padding: 15px 20px;
         display: flex;
         justify-content: space-between;
         align-items: center;
```

```
}

.products {
    display: grid;
    grid-template-columns: repeat(
        auto-fit,
        minmax(200px, 1fr)
    );
    gap: 20px;
    padding: 20px;
}

.card {
    background: white;
    border-radius: 8px;
    padding: 15px;
    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
    text-align: center;
}

.card img {
    width: 100%;
    height: 150px;
    object-fit: cover;
    border-radius: 4px;
}

.price {
    color: #E74C3C;
    font-size: 1.2em;
    font-weight: bold;
    margin: 8px 0;
}

.btn {
    background: #E74C3C;
    color: white;
    border: none;
    padding: 8px 16px;
    border-radius: 4px;
```

```
  cursor: pointer;
   width: 100%;
}

.btn:hover {
   background: #C0392B;
}

.cart-panel {
   position: fixed;
   right: 0;
   top: 60px;
   background: white;
   width: 300px;
   height: calc(100vh - 60px);
   box-shadow: -2px 0 8px rgba(0, 0, 0, 0.1);
   padding: 20px;
   overflow-y: auto;
   display: none;
}

.cart-panel.open {
   display: block;
}

#cartBadge {
   background: white;
   color: #E74C3C;
   border-radius: 50%;
   padding: 2px 6px;
   font-size: 12px;
   font-weight: bold;
}

#offlineBar {
   background: #FFA500;
   text-align: center;
   padding: 8px;
   display: none;
}
```

```
  </style>
</head>

<body>

  <!-- Offline Message -->
  <div id="offlineBar">
    ⚠ You are offline. Cart saved locally.
  </div>

  <!-- Header -->
  <header>
    <h2>□ ShopPWA</h2>

    <button
      onclick="toggleCart()"
      style="
        background: transparent;
        border: none;
        color: white;
        font-size: 20px;
        cursor: pointer;
      "
    >
      □ Cart
      <span id="cartBadge">0</span>
    </button>
  </header>

  <!-- Product List -->
  <div class="products" id="productList"></div>

  <!-- Cart Panel -->
  <div class="cart-panel" id="cartPanel">

    <h3>Your Cart</h3>

    <div id="cartItems"></div>

    <hr>

    <p>
      Total:
      <strong id="cartTotal">₹0</strong>
    </p>
```

```
    <button class="btn" onclick="checkout()">
      Checkout
    </button>

    <button
      onclick="toggleCart()"
      style="margin-top: 10px; width: 100%;"
    >
      Close
    </button>

  </div>

  <script>

    // Product List
    const products = [
      {
        id: 1,
        name: "Laptop",
        price: 55000,
        emoji: "□"
      },
      {
        id: 2,
        name: "Headphones",
        price: 2500,
        emoji: "□"
      },
      {
        id: 3,
        name: "Smartphone",
        price: 18000,
        emoji: "□"
      },
      {
        id: 4,
        name: "Tablet",
        price: 22000,
        emoji: "□"
      },
      {
        id: 5,
        name: "Keyboard",
        price: 1500,
        emoji: "⌨□
```

```
},
        {
            id: 6,
            name: "Mouse",
            price: 800,
            emoji: "□"
        }
    ];

    // Load cart from Local Storage
    let cart = JSON.parse(
        localStorage.getItem("pwa-cart") || "[]"
    );

    // Display Products
    function renderProducts() {

        document.getElementById("productList").innerHTML =
            products.map(p => `

                <div class="card">

                    <div style="font-size: 80px">
                        ${p.emoji}
                    </div>

                    <h3>${p.name}</h3>

                    <div class="price">
                        ₹${p.price.toLocaleString()}
                    </div>

                    <button
                        class="btn"
                        onclick="addToCart(${p.id})"
                    >
                        Add to Cart
                    </button>

                </div>

            `).join("");
    }

    // Add Product to Cart
    function addToCart(productId)
```

```
        const product = products.find(
            p => p.id === productId
        );

        const existing = cart.find(
            item => item.id === productId
        );

        if (existing) {
            existing.qty++;
        } else {
            cart.push({
                ...product,
                qty: 1
            });
        }

        saveCart();
        renderCart();

        alert(`${product.name} added to cart!`);
    }

    // Save Cart
    function saveCart() {

        localStorage.setItem(
            "pwa-cart",
            JSON.stringify(cart)
        );
    }

    // Display Cart
    function renderCart() {

        const total = cart.reduce(
            (sum, item) =>
                sum + item.price * item.qty,
            0
        );

        // Cart Badge
        document.getElementById("cartBadge").textContent =
            cart.reduce(
                (sum, item) => sum + item.qty,
);
```

```
        // Cart Total
        document.getElementById("cartTotal").textContent =
          `₹${total.toLocaleString()}`;

        // Cart Items
        document.getElementById("cartItems").innerHTML =
          cart.map(item => `

            <div
              style="
                display: flex;
                justify-content: space-between;
                margin: 8px 0;
              "
            >

              <span>
                ${item.emoji}
                ${item.name} x${item.qty}
              </span>

              <span>
                ₹${(
                  item.price * item.qty
                ).toLocaleString()}
              </span>

            </div>

          `).join("") || "<p>Cart is empty</p>";
      }

      // Open / Close Cart
      function toggleCart() {

        document
          .getElementById("cartPanel")
          .classList.toggle("open");
      }

      // Checkout
      function checkout() {
 if (!navigator.onLine)
          alert(
            "You are offline. Order will be placed when you reconnect."
```

```
            );

        } else {

            alert(
                "Order placed successfully! Thank you!"
            );

            cart = [];

            saveCart();
            renderCart();
        }
    }

    // Offline Event
    window.addEventListener("offline", () => {

        document.getElementById(
            "offlineBar"
        ).style.display = "block";

    });

    // Online Event
    window.addEventListener("online", () => {

        document.getElementById(
            "offlineBar"
        ).style.display = "none";

    });

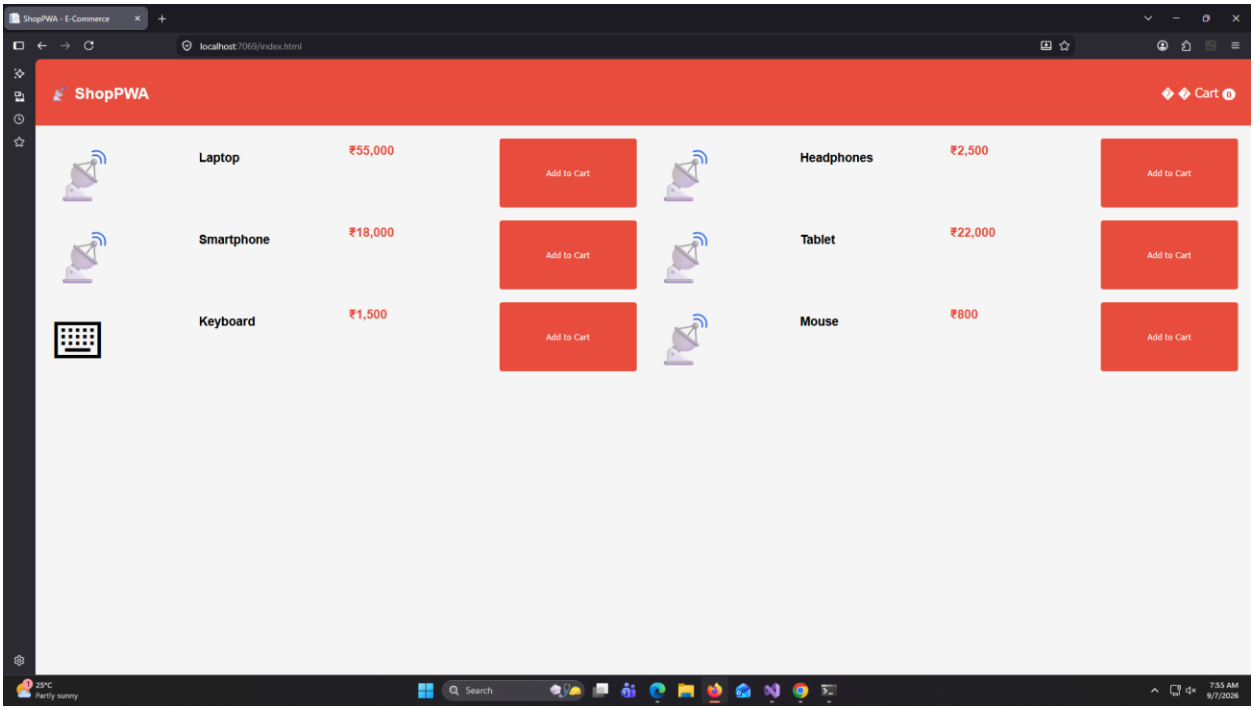
    // Initial Display
    renderProducts();
    renderCart();

    // Register Service Worker
    if ("serviceWorker" in navigator) {

        navigator.serviceWorker.register("/sw.js");

  </script>
</body>
</html>
```

Output


ShopPWA - E-Commerce
localhost:7069/index.html
ShopPWA
Cart 0
Laptop
₹55,000
Add to Cart
Headphones
₹2,500
Add to Cart
Smartphone
₹18,000
Add to Cart
Tablet
₹22,000
Add to Cart
Keyboard
₹1,500
Add to Cart
Mouse
₹800
Add to Cart
25°C
Partly sunny
Search
7:55 AM
9/7/2026